AI Concept Visualizer

Isabela Yepes, Manasvi Goyal, Nico Fidalgo

40-60%

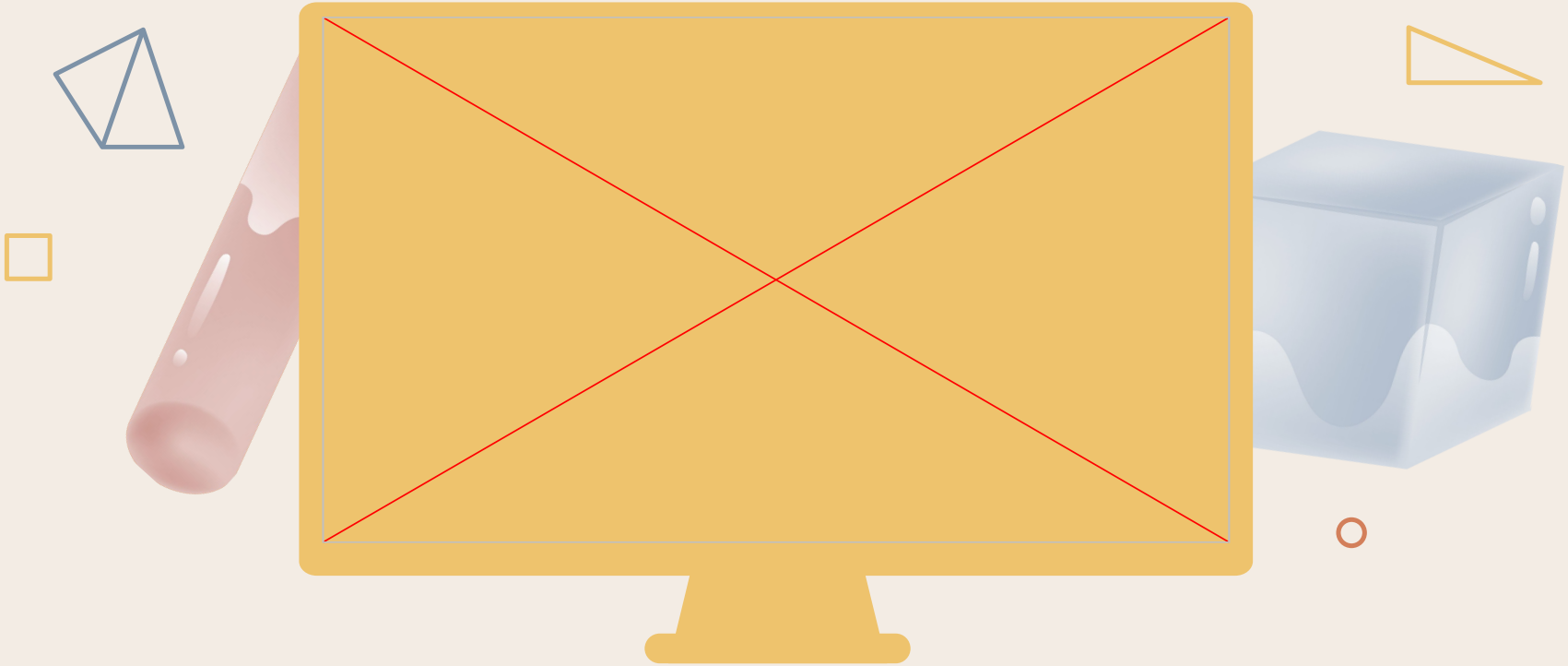
Of Students nationwide switched from STEM majors due to poor teaching. (Source)

29-42%

More information is retained by students taught with visual tools compared to text alone (Source)

89%

more likely to stay engaged and focused during lessons with visual materials, leading to higher persistence in the subject matter. (Source)

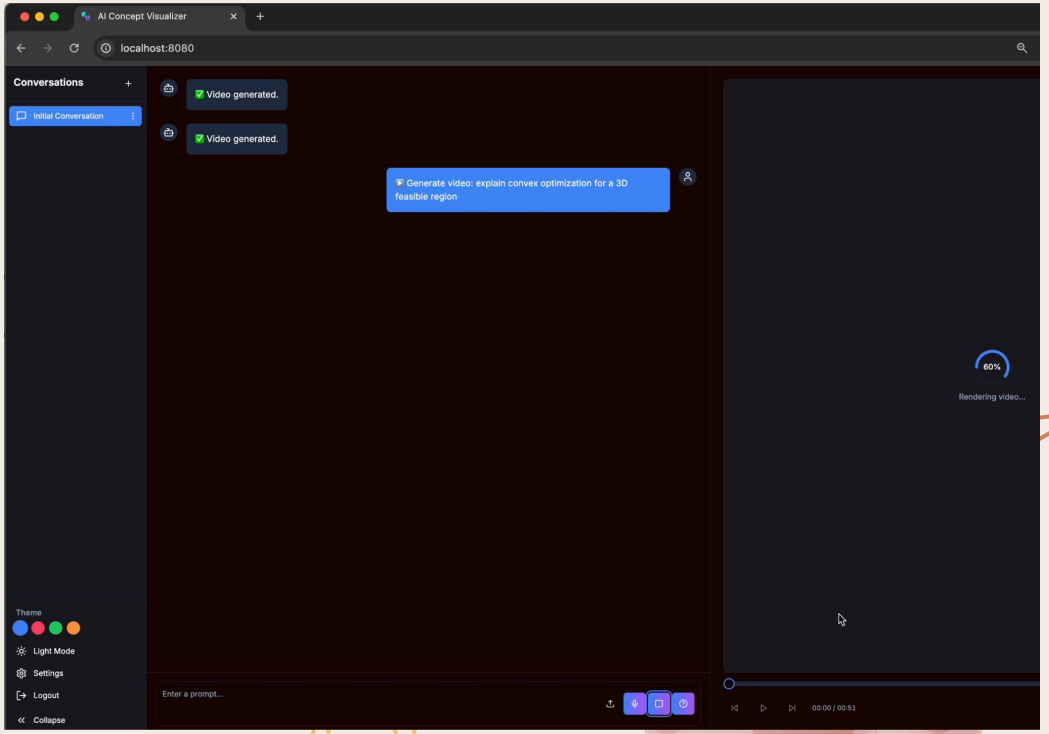


“Explain Pi”

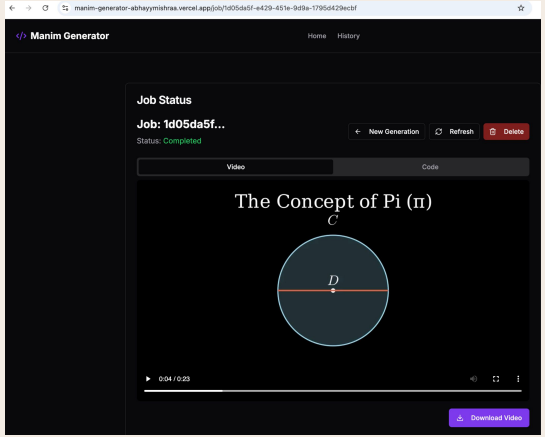
“Explain convex optimization for a 3D feasible region”

Our Solution

An application that automatically generates short-form, visually engaging explainers for technical concepts using generative AI.

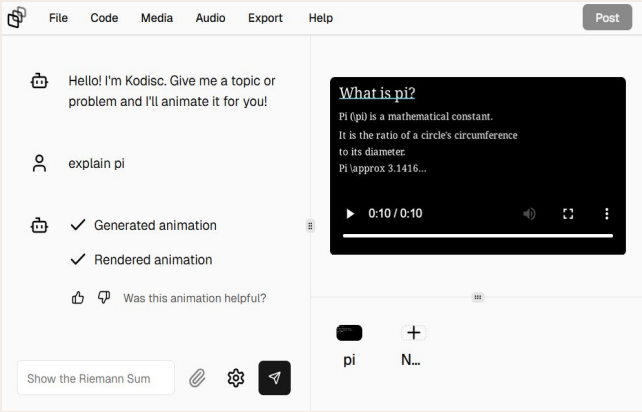


Competitors



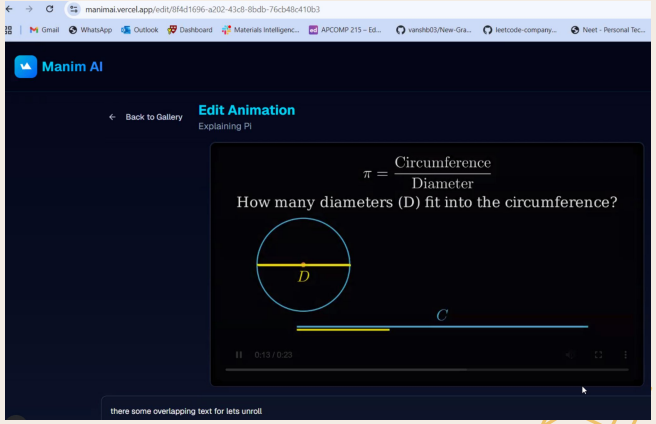
Manim Generator

- Pro: free
- Con: no audio



Kodisc

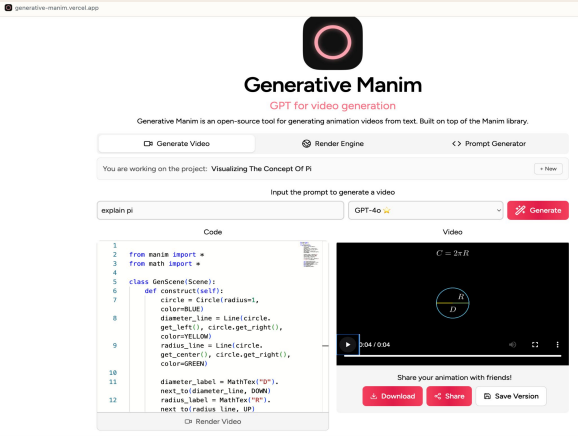
- Pro: free trial, quick generation
- Con: no audio, shorter videos



Manim AI

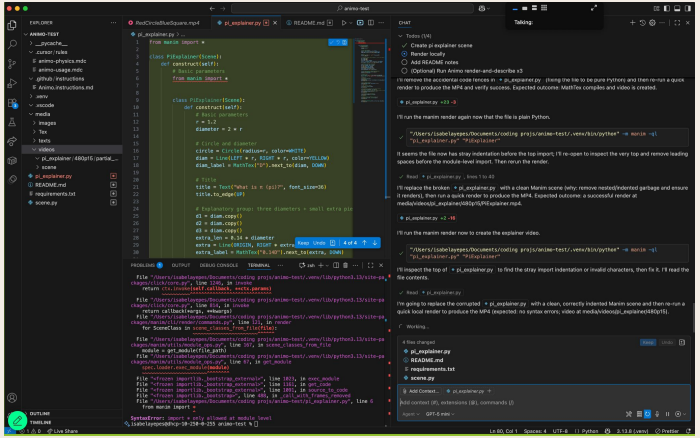
- Pro: good pedagogy and visuals, editable
- Con: no audio, edit did not work

Competitors



Generative-manim

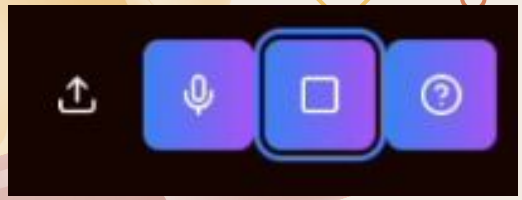
- Pro: free, no coding xp
- Con: short video, no audio



Animo

- Con: tedious debugging with copilot, not kid friendly
- Neutral: charges for RAG and AI feedback, uses local model

What we are offering



- A learning community for students and educators.
- Multilingual
- Audio with the visual explanations.
- Subtitles for Accessibility
- Generate Google Form Quizzes (MCP)
- Podcast (MCP)
- Retry in case of code errors (RAG)

Scalability and Efficiency

Technical Scalability:

- Autoscaling to spin up additional processing nodes during peak demand
- Asynchronous Processing using Google Cloud, to handle multiple generation requests concurrently

Performance Optimization:

- Prompt Engineering
- Deterministic sanitization fixes for complex visuals (code samples, 3D, e

Infrastructure Considerations:

- Google Cloud Platform for deployment and storage
- Docker for containerization

Further Development and Growth

PostGREs DB and Docker Containers (local), frontend on Vercel

MS4

11/25

12/10

MS5

Scale and Deploy (Kubernetes, Palumi), CI/CD

Video sharing platform

Community

?

?

Clients

Schools, individuals, etc.

?

Open source models, pricing

Cost

Prompt: Can you generate a manim animated video code script with manim voiceover gtts for teaching me about PPO RL?

Model	Can it Code?	Does the code run first time	Does the voiceover work	Quality (only if it could be debugged easily)	Open source?
Gemini Canvas	Yes	Yes	Yes	Best	No
Gemini Chat	Yes	No	Sometimes	Very Good	No
Llama 4 Maverick 17b	Yes	Yes	No	Ok (no visuals)	Yes (groq)
gpt-oss:120b	Yes	No	-	-	Yes (groq)
Claude	Yes	No	Sometimes	Good	No
Grok	Yes	No	Yes	Ok (visuals but overlap)	No
deepseekr1	No	No	No	-	Yes (ollama, groq)
gpt-oss:120b	Yes	No	-	-(outdated Manim doc error)	Yes (ollama, groq)
qwen3:32b	Yes	Yes	No, had to do it separately then worked but not synchronized	Not good (overlap)	Yes (groq)
gpt5	Yes	No	-	Code generation too slow	-